SYED ASJAD ABBAS

AI/ML Engineer | Applied AI, RAG & LLM Systems | Backend Engineering

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SUMMARY

AI/ML Engineer who builds LLM, RAG and multimodal systems on production-grade backends. Built a multi-tenant RAG platform gated in CI by a 50-case retrieval benchmark (Recall@5 0.97, MRR 0.85), a multimodal interview evaluator covered by 113 automated tests, and a video streaming API load-tested to 650 req/s at 500 concurrent users. Works across Python, FastAPI, PostgreSQL, Redis and React.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL, Dart

AI / ML & NLP: LLM Integration (Gemini, OpenAI), RAG, Embeddings, Hybrid Search, Reciprocal Rank Fusion, Cross-Encoder Reranking, RAG Evaluation (Recall@k, MRR, nDCG), DistilBERT, Computer Vision (YuNet, FERPlus, ONNX), Speech-to-Text, scikit-learn, Pandas, NumPy

Backend & APIs: FastAPI, Node.js, Express, REST API Design, JWT Authentication, RBAC, Multi-Tenancy, Background Workers, Idempotency, SQLAlchemy, Alembic, Prisma

Data & Infrastructure: PostgreSQL, pgvector, MongoDB, Redis, Docker, GitHub Actions (CI), Vercel, Git, Linux

Frontend & Testing: React, Next.js, Tailwind CSS, Flutter; Automated Testing, k6 Load Testing, Query Optimization (EXPLAIN ANALYZE)

PROFESSIONAL EXPERIENCE

AI/ML Engineer, SVYPE | Lahore (Hybrid) — Jul 2026 – Present

- Develop and integrate AI capabilities, including LLM and ML model integration, data preparation and evaluation, into internal platforms and client-facing products through Python backend services and APIs.
- Integrate LLMs and machine learning models into existing products through Python backend services and APIs, connecting AI functionality with the software systems that deliver it to users.

Full Stack Development Intern, SVYPE | Remote — May 2026 – Jul 2026

- Built ZePlay (see Projects) and contributed to TuneBox (Flutter); promoted to AI/ML Engineer on completion.

Founder & Lead Engineer, Zeploy Tech | Lahore (Remote) — Apr 2026 – Present

- Lead requirements, architecture, implementation and delivery of web, backend and AI products for clients, with 20+ projects delivered from requirements to launch.
- **Serene Management** (hotel PMS, Next.js, TypeScript, Prisma, PostgreSQL): 127 data models and 12 staff roles; idempotency keys and a SKIP LOCKED job worker prevent double billing. Cut search from 8.3 s to 0.45 s and sustained 50 req/s with 0 errors at P95 192 ms in single-instance load tests.
- **FlightOne** (AI travel platform) and **Zee** (Gemini assistant on zeploy.tech): LLM extracts typed trip parameters while the server computes every fare, so the model never generates prices; streaming responses with automatic fallback to a backup model.

Co-Founder & COO, 100 Se Support | Khairpur (Hybrid) — Jan 2021 – Present

- Run operations for a community welfare initiative: request verification, volunteers and accountable records.

SELECTED PROJECTS

Intelligence OS | Multi-tenant RAG platform | FastAPI, PostgreSQL + pgvector, Redis, Next.js, GitHub Actions — 2026

- Built hybrid retrieval combining pgvector HNSW vector search and PostgreSQL full-text search (GIN), merged with Reciprocal Rank Fusion, query rewriting and cross-encoder reranking; answers are grounded with page-level citations and tenant isolation enforced from the JWT.
- Built an evaluation framework and CI regression gate over a 50-case golden benchmark: Recall@5 0.97, MRR 0.85, nDCG@5 0.87, citation precision 1.00 and 100% correct refusal of unanswerable queries (offline deterministic judge); async evaluation runs support cancellation and crash recovery.

MockAI | Multimodal AI interview evaluation (Final Year Project) | FastAPI, React, MongoDB, FFmpeg — 2026

- Scores each recorded answer on content (DistilBERT semantic similarity), delivery (Google Speech-to-Text; pace, pauses, fillers) and composure (YuNet + FERPlus ONNX), fused 50/30/20 into an explainable report with coaching insights.
- Background evaluation worker with graceful degradation when the camera or microphone is missing and difficulty-weighted scoring; 113 automated tests across 10 suites, with full multimodal evaluation in about 36 s under 250 MB peak memory in end-to-end acceptance testing.

ZePlay | Adaptive video streaming platform | React, TypeScript, FastAPI, PostgreSQL, Redis, FFmpeg, k6 — 2026

- Background FFmpeg transcoding to 1080p, 720p and 480p HLS renditions for adaptive-bitrate playback; JWT auth, profiles, watch history, search and recommendations over a 100,017-title catalogue.
- Raised throughput 7.7× (84 to 650 req/s) at 500 virtual users with 0% failures and P95 1.68 s via Redis caching, GZip and multi-worker tuning (local k6 benchmark); keyset pagination cut deep-page queries from ~295 ms to 0.27 ms; 59/59 backend tests passing.

EDUCATION & CERTIFICATIONS

BS Computer Science, COMSATS University Islamabad, Lahore Campus — 2023 – 2027 (Expected)

Coursework: Machine Learning, Computer Vision, NLP, Database Systems | **Certifications:** Deep Learning Specialization (DeepLearning.AI), Google AI Essentials (Google), 2026